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Exploratory analysis of relationship between trunk and pelvic motion and shoulder motion during throwing in American football quarterbacks

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Title: Exploratory analysis of relationship between trunk and pelvic motion and shoulder motion during throwing in American football quarterbacks

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Abstract

Background: American football quarterbacks (QBs) often sustain shoulder and elbow injuries from throwing. During the late cocking phase of pitching, decreased shoulder abduction and horizontal adduction angles at maximum shoulder external rotation (MER) contribute to these injuries. The relationship between incorrect shoulder positions at MER and the trunk and pelvic rotation angles at stride foot contact (SFC) during the early cocking phase has been analyzed. However, most studies focus on baseball pitchers rather than QBs. The relationship between shoulder kinematics at MER, and trunk and pelvis kinematics at SFC in QBs during throwing remains underexplored. The purpose of this study was to explore the relationship between trunk and pelvic kinematics at SFC and shoulder kinematics at MER in QBs under non-game situations.

Methods: Eleven male American football QBs participated, each throwing five balls over 30 yards with maximum effort. Throwing motions were recorded at 240 Hz using an iPad positioned at the participant's side. Kinematic data were analyzed using Pitch AI™ to calculate the trunk (flexion/extension and non-throwing/throwing side rotation), pelvic (non-throwing/throwing side rotation), and hip-shoulder separation angles at SFC, as well as the shoulder (horizontal adduction/abduction, abduction/adduction, and external/internal rotation) angles at MER. The hip-shoulder separation angle is the difference between the pelvic and trunk rotation angles, with a positive value indicating trunk rotation toward the throwing side relative to the pelvis. Although PitchAI does not completely match data obtained with a marker-based motion capture system, its validity for trunk and pelvic kinematics is considered sufficient. Relationships between trunk and pelvic angles at SFC and shoulder angles at MER were evaluated using Pearson's product moment correlation coefficient (r_p) or Spearman rank correlation coefficient (r_s).

Results: Trunk (r_p= 0.779, p=0.005), pelvic（r_s= 0.691, p=0.019), and the hip-shoulder

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separation (r_p= 0.667, p=0.025) angle at SFC were positively correlated with shoulder horizontal adduction at MER. Trunk (r_p= -0.635, p=0.036) and pelvic (r_s= -0.682, p=0.021) rotation angles at SFC were negatively correlated with shoulder abduction angle at MER.

Conclusion: During the throwing motion in QBs, shoulder abduction and horizontal adduction angles at MER appear related to trunk and pelvic rotation movements at SFC.

Level of evidence: Basic Science Study; Kinesiology

Keywords: shoulder kinematics; trunk rotation angle; pelvis rotation angle; hip-shoulder separation angle; quarterbacks; exploratory

Quarterbacks (QBs) in American football are prone to acute injuries, such as shoulder dislocation and clavicle fractures, as well as shoulder and elbow problems from repeated throwing motions.[6,13] Kelly et al[6] surveyed shoulder and elbow disorders in National Football League QBs, while Radel et al[13] studied QBs under 18 years old. Their findings revealed that QBs consistently experience throwing-related injuries, including rotator cuff tears, shoulder impingement syndrome, and ulnar collateral ligament injuries, which can result in pain and limitations in throwing, potentially affecting athletic performance.

The throwing motion of a quarterback has unique movement patterns compared with baseball pitchers.[7,17] Specifically, QBs have shorter strides, variable arm slot and release angles, make rapid decisions under pressure, and emphasize accuracy over ball speed. These characteristics may influence overall body movement during throwing.

Limited studies have analyzed the throwing motion of QBs.[5,14] Fleisig et al[5] used 3D motion analysis to examine QBs and reported a ball speed of 21 m/s, anterior shoulder force of approximately 350 N, and elbow varus torque of approximately 54 Nm, indicating significant shoulder and elbow loads generated similar to baseball pitching.[5] Takagi et al[16] reported that excessive shoulder horizontal abduction at maximum external rotation (MER) increases anterior force of the shoulder, and Matsuo and Fleisig[11] found that decreased shoulder

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abduction during the cocking phase increases elbow varus torque. These findings highlight the importance of controlling shoulder motion to reduce joint load.

Throwing is a whole-body movement that transfers energy from the lower to upper extremities.[1] Controlling movements of the trunk and pelvis, which link the lower and upper extremities, is necessary to reduce the load on shoulder and elbow.[9,15] In baseball pitching, early trunk rotation at stride foot contact (SFC) increases shoulder horizontal abduction and decreases shoulder abduction during the late cocking phase.[9] Sakata et al[15]also reported that young baseball players with throwing injuries had low pelvic rotation during pitching. These studies suggest that trunk and pelvic control is crucial for managing shoulder movement during pitching.

When a quarterback throws, trunk and pelvis rotation occurs from SFC to ball release.[8] Like baseball pitching, QB throwing involves the whole body, and controlling trunk and pelvic movements may influence shoulder joint motion; however, this relationship has not been previously examined. Furthermore, QBs throw under conditions different from baseball pitchers, and understanding their basic throwing motion requires observation in non-game situations without opposing player pressure.

The purpose of this study was to explore the basic throwing movements of QBs by examining the relationship between trunk and pelvic kinematics at SFC and shoulder kinematics at MER under non-game conditions. Based on findings from baseball pitching, we formulated an exploratory hypothesis that trunk and pelvic kinematics at SFC are related to shoulder kinematics at MER in QBs.

Materials and methods

Participants

This study included 11 QBs from a university American football team or a club team affiliated with the Japan American Football League Xcellence (age: 25.1±6.3 years; body mass:

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81.9±9.3kg; height: 179.1±5.3cm; dominant hand: right, n=11; experience: 10.3±5.7years; player level: college students n=5, club team players n=6). Recruitment was conducted through the university team associated with the authors' institution and a club team previously attended by one of the co-authors (S.A.). Participants with a history of shoulder or elbow surgery, those unable to throw at maximum effort due to shoulder or elbow pain, and those with pain in the lower back or lower extremities, were excluded. The study protocol was approved by Research Ethics Committee (study number: 2023-301-A), and all participants provided written informed consent prior to participation.

Throw analysis

Participants were allowed unlimited time for warm-up and stretching, before testing. Each participant threw five balls toward a 1.1×1.1 m target placed 30 yards away in an outdoor setting. Throws were performed at maximum effort using an official ball (LONGBOMB model; Wilson Sporting Goods Co., Chicago, IL, USA) weighing 400–425 g. Participants were verbally instructed to "throw the ball as hard, as fast, and as powerfully as possible." Maximum effort was based on the subjective judgment of the participants.

Throwing motion was analyzed using PitchAI (3Motion AI Inc, Oakville, Canada),[3] an application that converts two-dimensional video into three-dimensional motion data using neural networks, and applies a markerless motion-capture system. The neural network employed in PitchAI comprises 38 input nodes representing joint center coordinates derived from video data, one hidden layer with 100 nodes using rectified liner unit activation, and 159 output nodes with liner activation corresponding to corrected joint center positions. This model was trained to refine markerless motion-capture outputs by referencing marker-based joint center data, enabling accurate motion analysis from standard video.

The throwing motion was recorded using the rear camera of an iPad (Apple Inc., Cupertino, CA, USA) at the default 1× setting . To standardize shooting conditions and minimize parallax

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error, participant stood with their bodies rotated 90° to the target. The iPad was placed anteriorly (abdominal side) at a height of 1.25 m from the ground and 6.0 m from the participant (Figure 1). The built-in level function ensured that the device was parallel to the ground. The camera was mounted on a tripod to prevent panning and was set to 240 Hz frequency, 1080p resolution, and landscape mode. Pitch AI™ was used to determine trunk (flexion/extension and rotation), pelvis (rotation), and hip-shoulder separation angles at SFC during the early cocking phase, as well as shoulder angles (horizontal adduction/abduction, abduction/adduction, and external/internal rotation) at MER during the late cocking phase (Figure 2).[3,5]

PitchAI identifies anatomical landmarks including the seventh cervical vertebra, suprasternal notch, eighth thoracic vertebra, xiphoid process, bilateral anterior and posterior superior iliac spines, lateral aspects of the head, acromion, medial and lateral humeral epicondyles, ulnar and radial styloid processes, hands, femoral condyles, malleoli, heels, and toes. These landmarks define local coordinate systems. Shoulder and hip joint centers were estimated from the acromion and pelvic landmarks, respectively. The trunk and pelvis angles were calculated relative to the ground. Trunk flexion was measured as the angle between a perpendicular line to the ground and a line connecting the centers of the pelvis and the trunk The position in which the trunk is parallel to the perpendicular line to the ground was defined as 0º of trunk flexion/extension, and positive values were considered as trunk flexion positions. The trunk rotation angle was defined as the angle between the line connecting both shoulders and the line connecting the pivot foot to the target. When the line connecting both shoulders and the line connecting the supporting foot to the target were parallel, trunk rotation was considered 0°, and positive values were indicative of trunk rotation on the non-throwing side. The pelvic rotation angle was defined as the angle between the line connecting both hip joints and the line connecting the pivot foot to the target. Pelvic rotation was defined as 0° when the line connecting both hip joints and the line connecting the supporting foot to the target were

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parallel, and positive values were indicative of pelvic rotation on the non-throwing side. The hip-shoulder separation angle was defined as the difference between the pelvic and trunk rotation angles, with a positive hip-shoulder separation angle indicating trunk rotation toward the throwing side relative to the pelvis. The shoulder angle was defined as the relative angle between the trunk and the upper arm. Shoulder horizontal adduction and abduction angles were measured as the angle between the line connecting both shoulders and the upper arm. The position in which the line connecting both shoulders and the upper arm was parallel was defined as 0° shoulder horizontal adduction/abduction, and positive values were defined as shoulder horizontal adduction. The shoulder abduction angle was measured as the angle between the upper arm and the line connecting the shoulder and hip on the throwing side. The drooping position of the upper arm was defined as 0° of shoulder abduction/adduction, and a positive value was defined as shoulder abduction. The shoulder external and internal rotation angles were measured by taking a line perpendicular to the trunk as the basic axis and measuring the angle between this axis and the upper arm. A position in which the shoulder is abducted at 90º, the elbow is flexed at 90º, and the forearm is facing forward was defined as 0º of shoulder internal/external rotation, and positive values were defined as shoulder external rotation.

To date, no studies have validated joint angle measurements from PitchAI during QB throwing motions. However, Dobos et al[3] compared marker-based motion capture system with PitchAI for baseball pitching, and reported coefficients of determination (R^2) and root mean square errors (RMSE). At MER, shoulder horizontal adduction had an R^2 of 0.33 and RMSE of 11.6°, shoulder abduction had an R^2 of 0.58 and RMSE of 13.7°, and shoulder external rotation had an R^2 of 0.42 and RMSE of 11.6°. At SFC, trunk flexion showed an R^2 of 0.18 and RMSE of 7.0°, trunk rotation had an R^2 of 0.78 and RMSE of 8.6°, and pelvic rotation had an R^2 of 0.84 and RMSE of 4.0°. These results indicate high agreement for trunk and pelvic rotation at SFC, but moderate agreement and larger errors for shoulder joint angles at MER.

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Despite these limitations, PitchAI was adopted as a practical tool for outdoor motion analysis.

All recorded throwing data was managed on the PitchAI platform, and participants provided consent for data storage and analysis via this AI-based system.

Statistical analysis

Statistical analyses were performed using JMP Pro software, version 16 (SAS Institute Inc., Cary, NC, USA). Data normality was assessed with the Shapiro–Wilk test. Associations between trunk, pelvic, and hip-shoulder separation kinematics at SFC and shoulder kinematics at MER were analyzed using Pearson's product moment correlation coefficient (r_p) or Spearman rank correlation coefficient (r_s). Shoulder kinematics data served as the objective variable, while trunk, pelvic, and hip-shoulder separation data were used as explanatory variables. When both variables were normally distributed, Pearson's correlation was used; otherwise, Spearman's correlation was used. With an effect size of 0.50, significance level of 0.05, and a power of analysis 0.8, the required sample size was calculated as 29 participants. However, this number was not achieved at the time of recruitment. Statistical significance was set at $p < 0.05$.

Results

In this study, none of the players met the exclusion criteria; therefore, the throws of 11 QBs were analyzed. Among all measured variables, only the pelvic rotation angle did not exhibit normality; all others demonstrated normal distribution.

During the throwing motion, the trunk flexion angle was 4.4 ± 5.5°, trunk rotation angle was 4.4 ± 29.6°, pelvic rotation angle (median [quartile deviation]) was 29.7 (40.5°), hip-shoulder separation angle was 9.8 ± 11.2°, shoulder horizontal adduction angle at MER was -19.8 ± 9.1°, abduction angle was 98.9 ± 4.7°, and external rotation angle was 105.2 ± 18.3° (Table I). Table II summarizes the correlations between trunk, pelvis, and hip-shoulder separation angles at SFC and shoulder angles at MER. A positive correlation was observed between trunk (r_p= 0.779,

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p=0.005), pelvic (r_s= 0.691, p=0.019) rotation angles, and hip-shoulder separation (r_p= 0.667, p=0.025) angles at SFC and shoulder horizontal adduction angle at MER. A negative correlation was observed between trunk (r_p= -0.635, p=0.036) and pelvic (r_s= -0.682, p=0.021) rotation angles at SFC and shoulder abduction angle at MER. No other correlations were observed.

Discussion

Shoulder and elbow injuries can occur in QBs as a result of throwing.[6,13] In baseball pitching, increased shoulder horizontal abduction and decreased shoulder abduction angle during the late cocking phase have been associated with shoulder and elbow injuries.[11,16] Additionally, shoulder joint motion during throwing is known to be related to trunk and pelvic motion.[9,15] QB throwing mechanics differ from those of baseball pitchers in several ways,[17] including the use of a heavier ball, shorter stride length, and greater variability in arm slot and release angle. These characteristics may influence motor control during throwing. Nevertheless, QB throwing is also a whole-body motion,[8] and trunk and pelvic motion may be related to shoulder motion, similar to baseball pitching. However, no previous studies have examined the relationship between trunk/pelvic and shoulder kinematics in QB throwing. Furthermore, although QBs typically throw under defensive pressure,[7] observing their throwing motion in non-game situations is essential for understanding the fundamental mechanics. Therefore, in this study, we investigated the relationship between trunk and pelvic kinematics and shoulder kinematics during QB throwing under non-game situation.

A significant finding of this study was a positive correlation between the hip-shoulder separation angle at SFC and the horizontal adduction angle at MER (Figure 3). Additionally, trunk and pelvic rotation angles at SFC were negatively correlated with the shoulder abduction angle at MER (Figure 4). These preliminary results suggest that the relationship between shoulder kinematics and trunk/pelvic rotational motion during quarterback (QB)

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throwing under non-game conditions resembles the patterns observed in baseball pitching.

Previous studies have examined the relationship between trunk and pelvic rotation movements at the SFC and the risk of shoulder and elbow surgery, as well as their effects on shoulder load.[4,12] However, only a few reports have examined the relationship between trunk and pelvic rotation at the SFC and the horizontal adduction angle of the shoulder at MER in detail. Lin et al[9] used a three-dimensional motion analysis to investigate the relationship between the trunk rotation angle at SFC and the horizontal adduction angle of the shoulder during the late cocking phase in college and professional baseball players. Their findings revealed that players with trunk rotation angles < 25° toward the throwing side relative to the pelvis at SFC (early trunk rotation) had increased shoulder horizontal abduction angles during the late cocking phase.[9] This can be understood as the trunk rotating excessively toward the non-throwing side at SFC, which causes the throwing upper limb to be left behind, resulting in a larger shoulder horizontal abduction angle at MER. Similar to the results reported by Lin et al,[9] our findings revealed a correlation between the hip-shoulder separation angle at SFC and the shoulder horizontal adduction angle at MER. Based on this, to prevent a decrease in the horizontal adduction angle of the shoulder at MER, it may be effective for QBs to rotate the trunk relative to the pelvis toward the throwing side at SFC.

The shoulder abduction angle at MER has been associated with elbow varus torque, with lower shoulder abduction angle at MER tending to coincide with higher elbow varus torque.[11] Davis et al[2] analyzed the pitching motion of growing baseball players and reported that players with a larger trunk rotation angle to the non-throwing side at SFC tended to exhibit larger elbow varus torque. Lin et al[9] reported that baseball players with early trunk rotation at SFC was linked to lower shoulder abduction angles during the late cocking phase. Based on these reports, trunk rotation at SFC may affect shoulder abduction at MER during baseball pitching, and a similar phenomenon appeared to occur in the throwing motion of the QBs in

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this study. Manzi et al[10] used a three-dimensional motion analysis device to examine the relationship between the pelvic rotation angle at SFC and elbow varus torque, and found no clear association between two variables. Although, they did not examine the shoulder abduction angle and the details are unclear, in baseball pitching, the pelvic rotation angle at SFC may not be associated with the shoulder abduction angle at MER. In contrast, a relationship exists between the pelvic rotation angle at SFC and the shoulder abduction angle at MER, which may be a characteristic of QB throwing movements.

This study had some limitations. First, the sample size was small; given this small cohort, the statistical reliability of the findings is constrained. The potential for Type I errors (false positives) remains, and the reduced statistical power may have increased the likelihood of Type II errors (false negatives). These limitations necessitate cautious interpretation of the results. Moreover, the small sample size precluded the use of multiple regression analysis. Future research should aim to replicate these findings with a larger cohort to improve robustness and generalizability. Second, there is potential measurement error associated with the use of a markerless motion capture system. According to Dobos et al,[3] the trunk and pelvic rotation angles estimated by PitchAI during baseball pitching exhibit relatively small errors. However, these values do not fully align with those obtained from marker-based motion capture systems, and shoulder joint angles may exhibit errors exceeding 10 degrees. Our facility does not possess a marker-based motion capture system suitable for deployment on a football field. Therefore, the reliability of the data obtained from PitchAI for analyzing QB throwing mechanics has not been validated. As such, the interpretation of PitchAI-derived kinematic data in football-specific contexts should be approached with caution. Third, all throws were performed in a non-game situation. As a result, the applicability of the present findings to actual collision football environments may be limited. In real-game situations, QBs must make rapid decisions under defensive pressure while dynamically adjusting foot positioning and kinetic chain

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coordination in response to multiple receiver options. Therefore, the controlled environment used in this study may not fully capture the complexity and variability of in-game throwing mechanics. Fourth, the absence of kinetic data represents another limitation. Since PitchAI only estimates joint angles, kinetic data could not be analyzed in this study. However, kinetic parameters are essential for investigating the relationship between throwing mechanics and throwing-related injuries. Therefore, including kinetic data remains crucial for future research. Despite these limitations, the preliminary associations observed between shoulder kinematics and other joint kinematics during QB throwing may represent a key contribution of this study. In training and rehabilitation settings where movement is intentionally assessed, kinematic relationships observed under standardized conditions may serve as a reference for evaluating throwing mechanics.

Conclusion

This study investigated the relationship between trunk and pelvic angles and shoulder joint angles during QB throwing. There was a positive correlation between the hip-shoulder separation angle at SFC and the shoulder horizontal adduction angle at MER. Additionally, the trunk and pelvic rotation angle at SFC was negatively correlated with the shoulder abduction angle at MER. These results suggest that controlling the rotation of the trunk and pelvis during quarterback throwing may assist in optimizing mechanics.

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Figure 1. Measurement setup

The figure shows the setup of equipment and participant positioning during measurement.

Figure 2. Analytic phase of a throwing

A; Stride foot contact during the early cocking phase (SFC)

B; Shoulder maximum external rotation during late cocking phase (MER)

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Figure 3. Correlation between hip-shoulder separation angle at SFC and shoulder horizontal adduction angle at MER

Figure 4. Correlation between trunk and pelvic rotation at SFC and shoulder abduction angle at MER

Table I. Kinematic data of the trunk, pelvis, and hip-shoulder separation at SFC and the shoulder at MER

Data are presented as the mean ± standard deviation and medians and quartile deviation.

[] indicates quartile deviation

SFC, stride foot contact; MER, maximum external rotation

Table II. Correlation analysis between the trunk, pelvic, and hip–shoulder separation angles at SFC and shoulder angle at MER

SFC, stride foot contact; MER, maximum external rotation

Spearman's rank correlation coefficient was used for correlation analysis with pelvic rotation angle as the explanatory variable. Pearson's product moment correlation coefficient analysis was used for other variables.

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Table Ⅱ. Correlation analysis between the trunk, pelvic, and hip–shoulder separation angles at SFC and shoulder angle at MER

	Shoulder angle					
	Horizontal adduction		Abduction		External rotation	
	Correlation coefficient	P value	Correlation coefficient	P value	Correlation coefficient	P value
Trunk flexion	-0.527	0.096	0.502	0.115	-0.488	0.128
Trunk rotation	0.779	0.005	-0.635	0.036	0.498	0.119
Pelvic rotation	0.691	0.019	-0.682	0.021	0.591	0.056
Hip-Shoulder Separation	0.667	0.025	-0.500	0.118	0.574	0.065

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Table I. Kinematic data of the trunk, pelvis, and hip-shoulder separation at SFC and the shoulder at MER

	Variable (°)	Data
	Trunk	
SFC	Flexion (+)/Extension (-)	4.4±5.5
	Non-throwing side rotation (+)/ throwing side rotation (-)	4.4±29.6
	Pelvic non-throwing side rotation (+)/throwing side rotation (-)	29.7 (40.5)
	Hip-Shoulder separation	9.8±11.2
	Shoulder	
MER	Horizontal adduction (+)/Horizontal abduction (-)	-19.8±9.1
	Abduction (+)/Adduction (-)	98.9±4.7
	External rotation (+)/Internal rotation (-)	105.2±18.3

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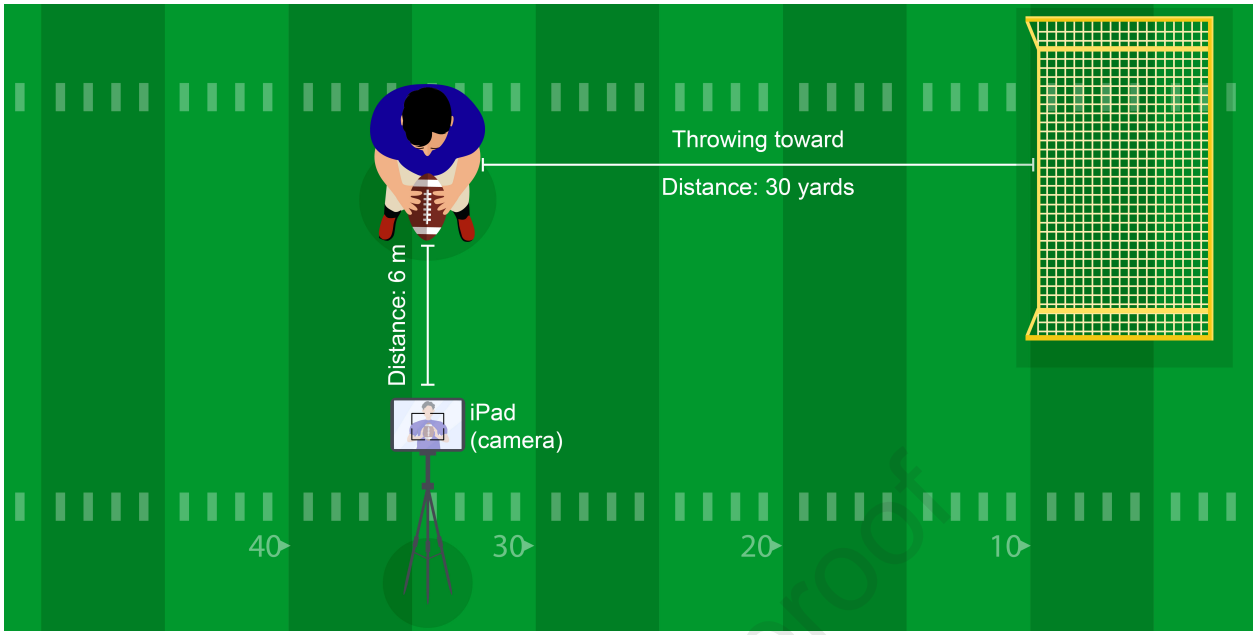

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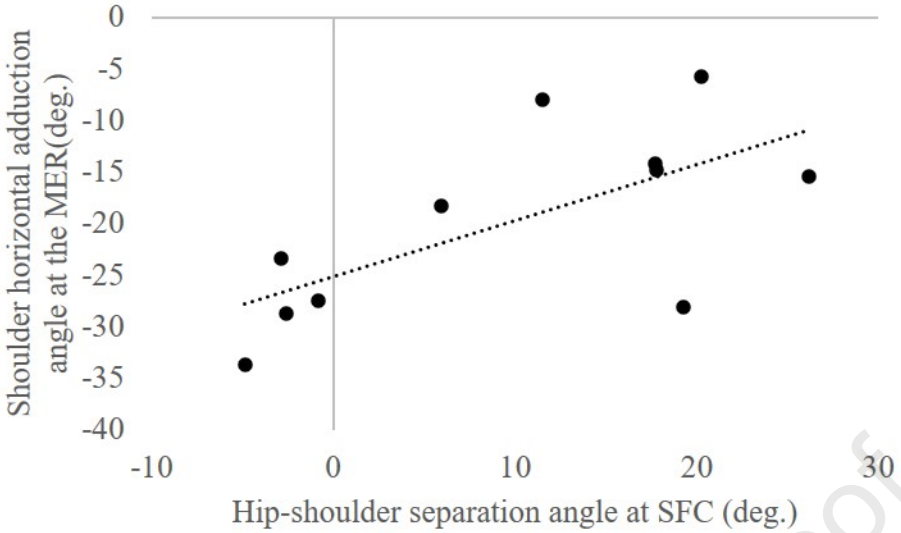

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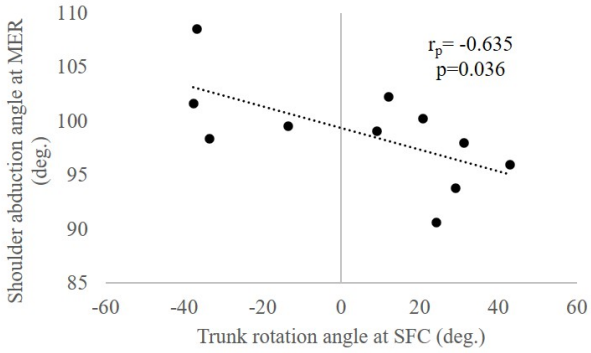

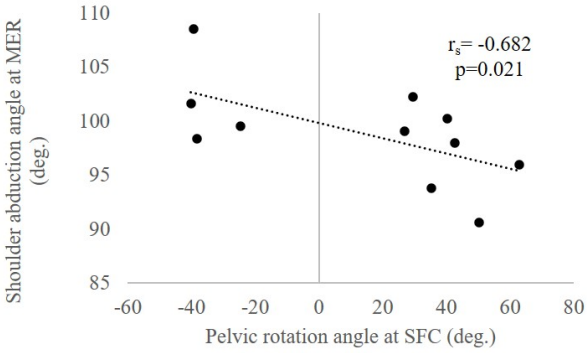

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the conflict of interest (CoI) statement

Title :

Relationship between trunk and pelvic motion and shoulder motion during throwing in American football quarterbacks

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